Telling Pills Apart

Sorting pills by shape, color and size with Python, OpenCV and PyTorch, using reference photos from the National Institutes of Health · Project report, November 2026

A Bergen County Youth Assembly project · Active Bergen (Health and Safety), directed by Justin Yoo · Model and analysis built by Alex Kim · Early development mentored by Hyunsoo Song, biostatistician, Johns Hopkins

8,784	4,231	95%	89%	98.5%
NIH reference photos used	different medicines	shape accuracy (ResNet18)	color accuracy (ResNet18)	of medicines have a look-alike

Part 1. What this project does

People who take several medicines a day, especially older adults and people with low vision, often tell their pills apart by how the pills look: this one is the small white round one, that one is the long yellow one. That works until two of the medicines look the same. I wanted to find out how well a computer can read the same three things a person reads (shape, color and size) from a photo, and how often two different medicines share all three.

The project has three pieces. First, I took 8,784 reference photos of pills from the National Library of Medicine and attached to each photo the official shape, color and size of that pill. Second, I trained two different kinds of model to predict shape and color from the photo alone, and compared them. Third, I used the same tables to measure how crowded the look-alike problem really is, and built a checker that compares two pill photos and rates the risk of confusing them as high, medium or low.

> **What the tool does not do.** It does not tell you which medicine a pill is. The data shows why: shape, color and size together are shared by dozens of medicines, and only the imprint (the letters and numbers pressed into the pill) separates them. So the finished tool is a warning tool, not an identifier. It says “these two look alike, keep them apart,” and leaves identification to the pharmacist and the label.

The checker

The look-alike checker takes two photos, runs the shape, color and size predictions on each, and adds up a risk score: 40 points if the shapes match, 40 if the colors match (with extra points if the exact tint is close), and 20 if the sizes are within about 1.5 mm. A score of 75 or more is reported as high risk, 45 to 74 as medium, and below 45 as low, together with the reasons in plain words (“both look round, both white, about the same size”).

Part 2. The data

All of the data comes from two public collections published by the National Library of Medicine, part of the National Institutes of Health. Both are free to download and are released for research and development use.

Source	What it provides	How I used it
C3PI / RxIMAGE (*rximage.zip*)	High-quality photos of prescription pills taken under laboratory lighting from directly above, front and back of each pill	The images. Each file is named by the pill's National Drug Code (NDC), which is how the photo is linked to its labels.
Pillbox (retired in 2021, data still public)	A table of 80,460 pills with the shape, color, size in millimeters, imprint and medicine name taken from the FDA label	The labels. Joined to each photo by the first nine digits of the NDC.

After the join, 8,784 photos had a usable file and 8,500 of them (97 percent) had a shape and color label. These photos cover 4,231 distinct medicines, so most medicines appear twice (front and back). I grouped rare shape names into eleven shape classes and used the first listed color for pills with more than one color, which gives eleven color classes.

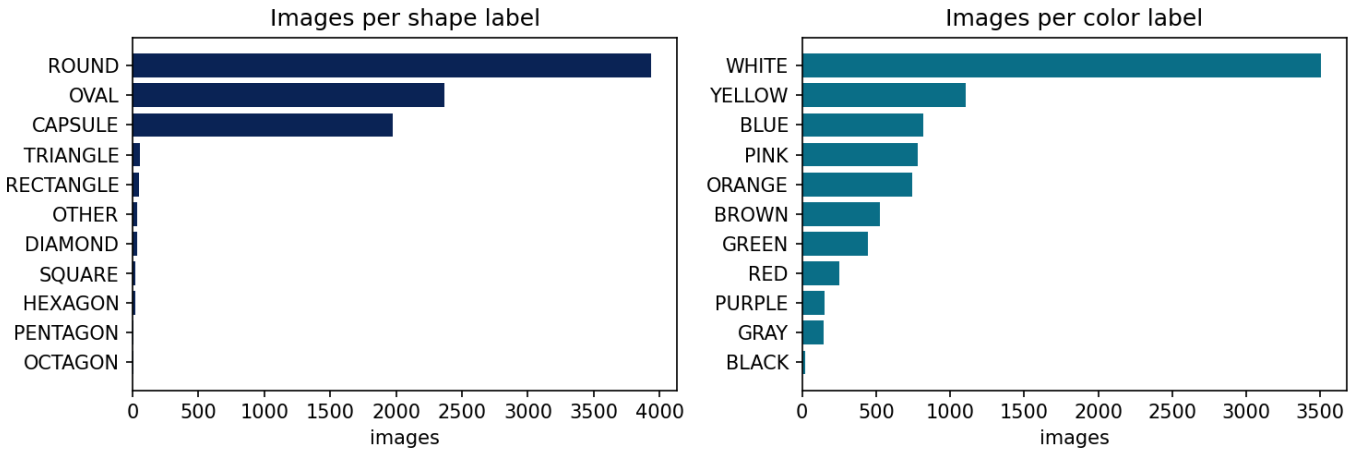


Figure 1. How the 8,500 labeled photos are distributed. Round, oval and capsule cover 97 percent of the shapes, and white alone is 41 percent of the colors.

This imbalance matters for reading the results. A model that is good at round white pills gets most of the test set right before it has learned anything about hexagons or gray pills, so I also looked at the mistakes class by class in Part 4.

Part 3. The two models

Model A: measurements plus a Random Forest

For each photo, OpenCV separates the pill from the background and then measures the pill in twenty numbers. For shape it records how round the outline is, how long the pill is compared with its width, how many corners a simplified outline has, and how much of its bounding box the pill fills. For color it records the average hue, saturation and brightness of the pixels inside the pill, plus the red, green and blue averages. For size it records the pill's width in pixels. Those twenty numbers go into a Random Forest, which is a collection of 400 decision trees that each vote on the shape or color. A separate forest predicts the size in millimeters.

The advantage of this model is that every input is a measurement I chose and can explain. After training, the forest also reports which measurements it relied on most. For shape the top five were: how much of the bounding box the pill fills (extent), roundness (circularity), the long-to-short axis ratio of a fitted ellipse, the aspect ratio of the bounding rectangle, and the pill's area in pixels.

Model B: a neural network with transfer learning

The second model is ResNet18, a convolutional neural network that was already trained on 1.2 million everyday photographs. I kept everything it had learned about edges, textures and colors and replaced only its final layer with two new "heads," one that predicts shape and one that predicts color. Both heads were trained at the same time in PyTorch on a free Colab GPU for 15 passes over the data (epochs), with random rotations and flips so the network does not learn that pills always point the same way.

How the test was run

For both models, 20 percent of the photos were set aside before training and never shown to the model. Every accuracy figure in this report is measured on those unseen photos. Model A was tested on 1,700 photos and Model B on 1,758; the small difference comes from the two scripts drawing their random 20 percent separately.

Part 4. Results

Model	Shape accuracy	Color accuracy	Test photos
A. Random Forest on OpenCV measurements	92.8%	85.2%	1,700
B. ResNet18 transfer learning, 15 epochs	**94.8%**	**88.6%**	1,758
A. Random Forest size estimate	average error 1.50 mm; 75% of pills within 2 mm of the true size		

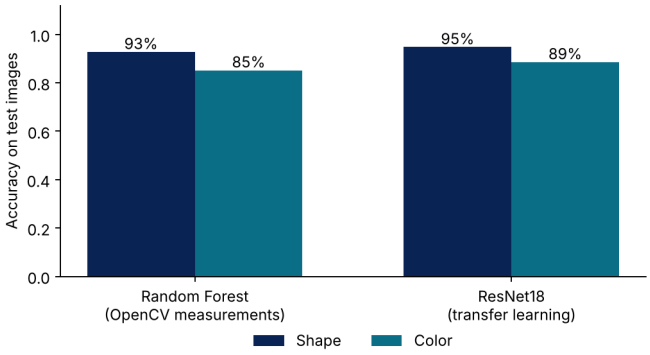


Figure 2. The two models side by side.

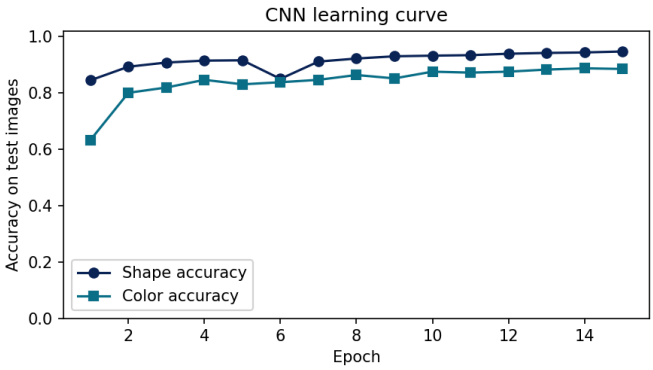


Figure 3. ResNet18 accuracy after each training pass. Shape went from 84.5% after one pass to 94.8% after fifteen.

The neural network wins, but by less than I expected: two points on shape and three on color. Twenty hand-chosen measurements get most of the way to what a network with eleven million parameters learns on its own. The learning curve was still rising slightly at epoch 15, so a longer run would likely add another point or two.

Where the models go wrong

The confusion matrices below show, for each true class (rows), what the model predicted (columns). Reading them tells more than the single accuracy number.

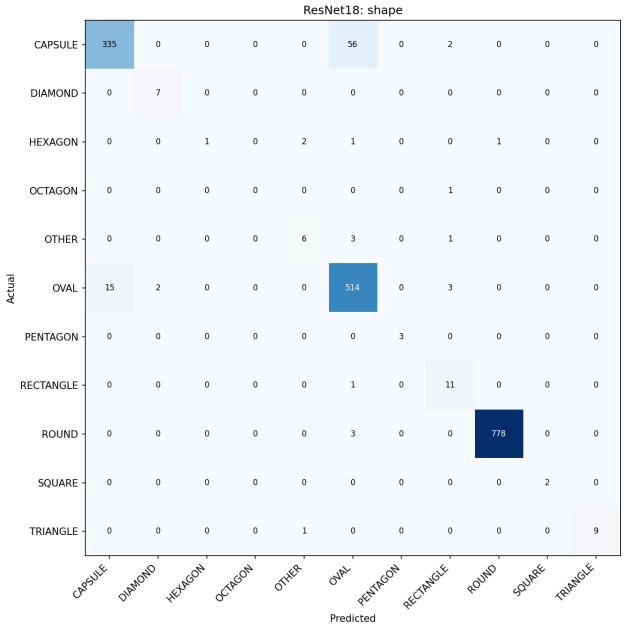


Figure 4. ResNet18, shape. Rows are the true shape, columns the predicted shape.

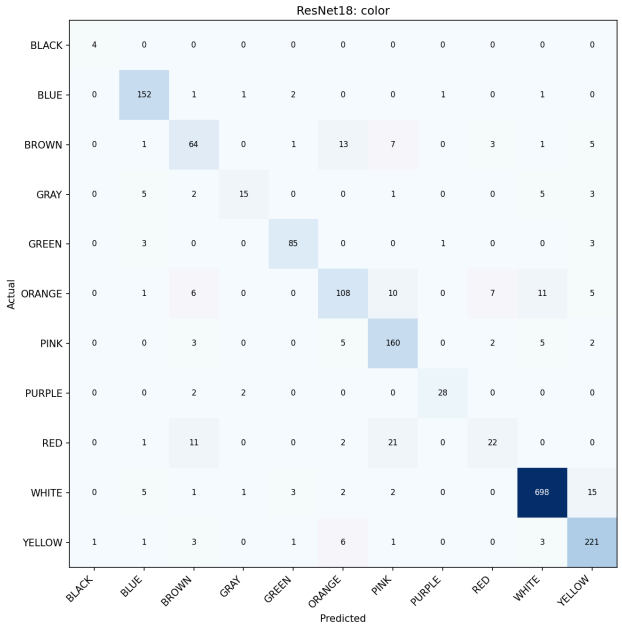


Figure 5. ResNet18, color.

- **Capsule versus oval is the main shape error.** ResNet18 called 56 capsules “oval,” and the Random Forest made the same mistake 53 times. Seen from above, a capsule and a long oval tablet have nearly the same outline; the difference is a seam and a slight thickness that a top-down photo barely shows. Round pills, by contrast, were almost never missed (778 of 781 for ResNet18).
- **Color errors sit on real boundaries.** Red was the weakest color for both models: ResNet18 called 21 of 57 red pills “pink” and 11 “brown.” Orange was split between pink, white and brown; brown leaked into orange. These are not random mistakes. They are the colors where the FDA label itself has to choose one word for a shade that sits between two.
- **White is easy and common.** ResNet18 got 698 of 727 white pills right. Because white is 41 percent of the data, this one class carries a large share of the overall color accuracy. On the rarer colors (gray, red, brown) both models are closer to 60 to 75 percent.
- **Both models fail in the same places.** That is the most useful finding of the comparison. If the errors came from the models, I would expect two very different methods to fail differently. They do not, which points to the labels and the photos as the limit, not the method.

Part 5. How crowded the look-alike problem is

Before any model runs, the Pillbox table alone answers the question that motivated the project. For every medicine with a full label I counted how many other medicines have the same shape and the same color and a size within one millimeter.

98.5%
of 4,089 medicines have at least one look-alike

87.8%
have ten or more

60
look-alikes for the typical (median) medicine

483
look-alikes for a round white 7 mm pill

Shape	Color	Size	Different medicines	Examples
round	white	6 mm	183	furosemide, dipyridamole, methimazole, benztropine
round	white	8 mm	171	rizatriptan, pindolol, promethazine
round	white	10 mm	137	prednisone, cortisone, metoprolol with hydrochlorothiazide
round	white	7 mm	130	minoxidil, acarbose, hydroxyzine, ondansetron
round	white	9 mm	86	prednisone, ciprofloxacin, sumatriptan
oval	white	19 mm	77	acyclovir, etodolac, oxaprozin
round	yellow	6 mm	58	repaglinide, trospium

Table 2. The most crowded shape, color and size buckets among the 4,089 photographed medicines with full labels. Size is rounded to the nearest millimeter. Across the full Pillbox table of 80,460 pills (not only the photographed ones), the share with at least one look-alike is 99.9 percent.

Some of these pairs matter medically. Prednisone (a steroid) and ciprofloxacin (an antibiotic) both appear as round white 9 mm tablets. Furosemide (a water pill) and methimazole (a thyroid medicine) both appear as round white 6 mm tablets. A person who keeps these in the same dish and relies on looks alone has no way to tell them apart.

This is what turned the project from “identify the pill” into “warn about the pair.” With these numbers, a model that names a medicine from its appearance would be wrong far more often than right, and confidently wrong is the worst thing a tool for older adults can be.

Part 6. Limits and what comes next

Limits

- **Lab photos only.** Every training and test photo was taken under controlled lighting from directly above on a plain background. A phone photo on a kitchen table has shadows, a tilted angle and a busy background. The accuracies in this report should be read as an upper bound for real use, not as a promise.
- **Size needs a reference.** The size model learned the pixel-to-millimeter relationship of the NIH camera setup. On a phone photo, a coin or ruler in the frame would be needed to make the millimeter estimate meaningful.
- **Labels are words, not measurements.** “Red,” “pink” and “orange” are names an FDA label writer chose. The model inherits every blurry boundary in those names.
- **Not a medical device.** The NIH releases these images for research. Anything built from them should say clearly that it is a study aid and that identification belongs to the pharmacist and the label.

Next steps

- Test on phone photos. The same NIH collection includes 43,544 consumer-style photos (the C3PI_Test set). Training on the lab photos and testing on those is the honest version of this experiment, and the code already supports it.
- Train longer and merge the capsule and oval classes for the warning tool, since the checker only needs to know that two pills share an outline.
- Put the checker behind a simple page where a user can upload two photos and read the result in large type.

What was built

The project is a Python package with seven scripts (download, index, measure, train the forest, train the network, look-alike analysis, report) and a Colab notebook that runs them in order. Libraries: OpenCV, pandas, scikit-learn, PyTorch and torchvision, matplotlib. Training ran on a free Colab T4 GPU; the Random Forest ran on the CPU in a few minutes and the network in about 30 minutes.

Data: National Library of Medicine, Computational Photography Project for Pill Identification (C3PI), *rximage.zip*, and the Pillbox dataset (retired January 28, 2021). Both are released for research and development use.